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17. Two resistors of resistances R_1 = 150 ± 2 Ohm and R_2 = 220 ± 6 Ohm are connected in parallel combination. Calculate the equivalent resistance.

Hint: $\frac{1}{R_1} = \frac{1}{R_1} + \frac{1}{R_2}$

Solution:

The equivalent resistance of a parallel combination

$$R' = \frac{R_1 R_2}{R_1 + R_2}$$

$$= \frac{150 \times 220}{150 + 220} = \frac{33000}{370} = 89.1\ Ohm$$

We know that, $\frac{1}{R'} = \frac{1}{R_1} + \frac{1}{R_2}$

$$\frac{\Delta R'}{(R')^2} = \frac{\Delta R_1}{R_1^{\,2}} + \frac{\Delta R_2}{R_2^{\,2}}$$

$$\Delta R' = (R')^2 \frac{\Delta R_1}{R_1^{\,2}} + (R')^2 \frac{\Delta R_2}{R_2^{\,2}}$$

$$= \left(\frac{R'}{R_1}\right)^2 \Delta R_1 + \left(\frac{R'}{R_2}\right)^2 \Delta R_2$$

Substituting the value,

$$\Delta R' = \left[\frac{89.1}{150}\right]^2 \times 2 + \left[\frac{89.1}{220}\right]^2 \times 6$$

$$= 0.070 + 0.098 = 0.168$$

R' = 89.1 ±0.168 Ohm.

18. A capacitor of capacitance C = $3.0 \pm 0.1 \mu F$ is charged to a voltage of V = $18 \pm 0.4 Volt$. Calculate the charge Q [Use Q= CV]

Solution:

$(C + \Delta C) = (3.0 \pm 0.1)\ \mu f$

$(V + \Delta V) = (18 \pm 0.4)\ V$

$Q = CV$

$Q = 3.0 \times 10^{-6} \times 18 = 54 \times 10^{-6}$ coulomb

$$\text{Error in C} = \frac{\Delta C}{C} \times 100$$

$$= \frac{0.1}{3} \times 100 = 3.3\%$$

$$\text{Error in V} = \frac{\Delta V}{V} \times 100$$

$$= \frac{0.4}{18} \times 100 = 2.2\%$$

Error in Q = Error in C + Error in V

= 3.3% + 2.2% =5.5%

∴ Charge Q = $(54 \times 10^{-6} \pm 5.5\%)$ C

SOLVED EXAMPLE UNIT-2

1. The position vector for a particle is represented be $\vec{r} = 3t^2\hat{i} + 5t\hat{j} + 6\hat{k}$, find the velocity and speed of the particle at t = 3 sec?

Solution:

$$\vec{v} = \frac{\vec{dr}}{dt} = 6t\hat{i} + 5\,\hat{j}.$$

The velocity at any time '*t*' is given by $\vec{v} = 6t\hat{i} + 5\,\hat{j}.$

The magnitude of velocity is speed. The speed at any time '*t*' is then given by